

CYBER SECURITY PROJECTS

Project Title: Vulnerability Scanner Tool

Covers modules: Network Scanning, Python Scripting, Port Analysis

Tasks

- Scan open ports on target systems
- Detect common vulnerabilities
- Generate scan reports
- Implement basic threat alerts

Outcome: Demonstrates network security analysis + vulnerability assessment skills.

Project Title: Phishing Detection System

Covers modules: URL Analysis, Machine Learning Basics, Cyber Threat Detection

Tasks

- Analyze suspicious URLs
- Detect phishing patterns
- Build blacklist/whitelist system
- Generate risk score for websites

Outcome: Shows cybersecurity automation + phishing attack prevention techniques.

Project Title: Secure File Encryption System

Covers modules: Cryptography, AES Encryption, File Security

Tasks

- Encrypt and decrypt files securely
- Implement user authentication
- Generate encryption keys
- Build secure file access system

Outcome: Demonstrates data protection + cryptographic implementation skills.

Project Title: Password Strength Analyzer

Covers modules: Data Cryptography, Secure Identity Management, Python Systems Engineering

Tasks

- Check password complexity
- Detect weak password patterns
- Suggest secure password improvements
- Implement password hashing techniques

Outcome: Demonstrates authentication security + secure credential handling.